Stat 414 Notes

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Course Goals for STAT 414

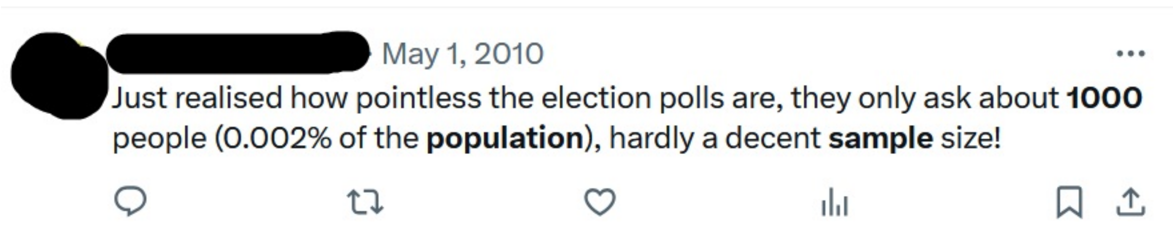
STAT 414 is an introduction to sampling. By the end of the course, you will:

- Learn the basic methodology of how classical survey samples are conducted based on simple random sampling.
- Recognize sampling and non-sampling errors.
- Be able to use extra information in sampling via ratio estimators and regression estimators.
- Be able to define and use stratified samples and know why they are often better than SRS.
- Be able to design one and two stage cluster samples and know their key properties.
- Recognize strengths and weaknesses of different sampling approaches.
- Use standard R packages to analyze survey data and estimate population means, proportions, totals, and more.

1 Introduction to Sampling

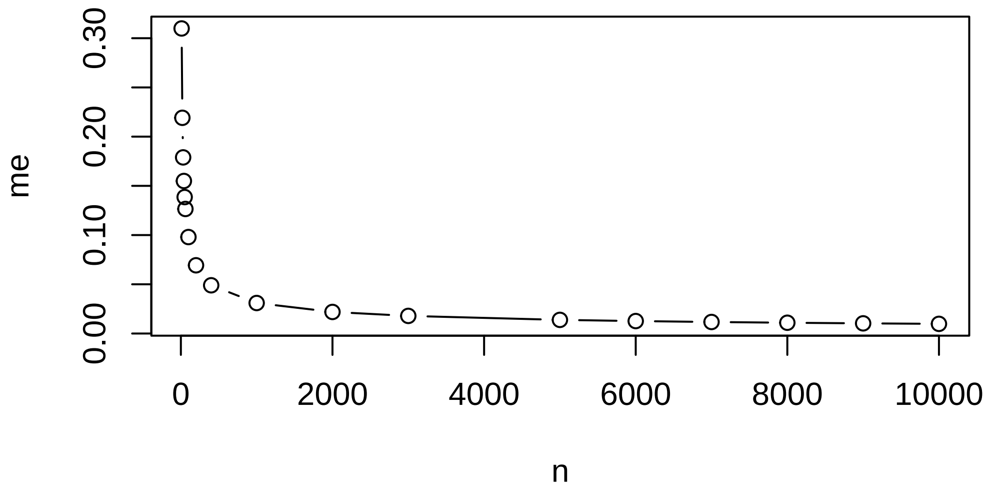
1.1 Introduction

If you spend any time at all on social media (especially in an election year), you're likely to come across a post like this:



What do you think? Do you agree?

```
n<-c(10,20,30,40,50,60,100,200,400,1000,2000,3000,5000,6000,7000,8000,9000,10000)
me<-1.96*sqrt(0.25/n)
plot(n,me,type="b")
```

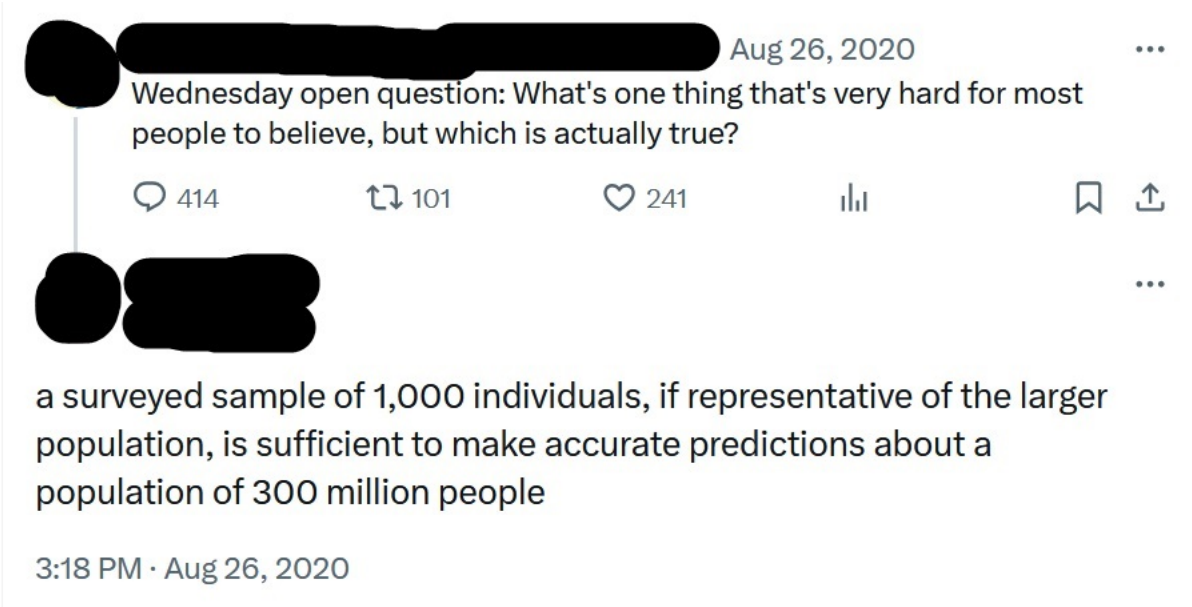


So why even sample?

1. Cheaper

2. Faster

3. **More Accurate** (?!?!?)



[redacted] Aug 26, 2020

Wednesday open question: What's one thing that's very hard for most people to believe, but which is actually true?

414 101 241

[redacted]

a surveyed sample of 1,000 individuals, if representative of the larger population, is sufficient to make accurate predictions about a population of 300 million people

3:18 PM · Aug 26, 2020

What's key here?

What we will spend this semester studying is how to ensure this, to make sure we can make accurate predictions. This will involve answering questions focused on:

- Survey design
- Preparation
- Collect and prepare data
- Data analysis

How is sampling different from previous courses, like STAT 212?

How is sampling different from previous courses, like STAT 262 or STAT 380?

How is sampling different from previous courses, like STAT 102 or STAT 318?

However, like previous courses, sample surveys do have an overarching process, some of which overlap with steps/processes we've seen before. We'll go through the principal considerations in a sample survey, along with some definitions and (possibly) new notation.

1.2 Considerations in planning and executing a survey

1. **Objectives of the survey**
2. **Defining the population** In sampling, the word **population** needs some clarification. Generally, it means the larger set from which the sample is chosen, and it's the larger group we wish to make inference/draw conclusions about. And a **sample** is a subset of the population. However, we must distinguish between

- **Target population**

- **Sampled population**

Ideally, these two overlap completely, but this does not always occur. Plus, we also need to consider the **sampling frame population**

I love the figure on page 5 of the book:

Within the **sampling frame** we have

- **Observation unit**

- **Sampling unit**

For example,

3. **Identifying potential sources of bias and ameliorating them**: In previous classes, like STAT 262 or STAT 380, we use the word **bias** to mean an estimator that systematically over- or under-estimates the true population characteristic. In sample surveys, we still use the same definition, but now biased estimators typically arise for two reasons:

   - Population characteristic of interest (Estimation bias):

   - Selection and/or measurement bias: Selection bias occurs when there is a mismatch between the sampled population and the target population. Measurement bias occurs when the survey observation tends to differ from the truth in a particular direction.

We'll talk about each of these in more detail.

Selection bias

- Convenience samples

- Judgment samples

- Self-selected samples

- Undercoverage

- Overcoverage

- Nonresponse

Measurement bias

Why can it occur?

- Unavoidable Reasons

- People!

- Survey mode: method used to distribute and gather responses

So we're concerned about response bias–what now? Careful question design is key. We could spend weeks on just questionnaire design, but will only do a very brief overview. There are other resources on campus that can help with this aspect of surveys (Bureau of Sociological Research).

We'll start with types of questions to avoid, and then move on to other considerations when writing questions.

Common types of biased questions:

- Leading questions

  How to fix it:

- Assumptive questions

  How to fix it:

- Double-barreled questions

How to fix it:

- Double negatives

How to fix it:

Other considerations:

- Pay careful attention to answer scale options

- Check formatting

- Avoid jargon, and be specific

- Open or closed questions?

- Question order matters

Probably most important, test questions ahead of time with the target population.

4. **Choosing the sampling design/data collection approach**

5. **Choosing the analysis approach**

6. **Data analysis**

7. **What's next?** The more information we have about a population, the easier it is to devise a sampling plan that will give accurate estimates. Any past survey can serve as a guide for future samples, as it will provide information about variablity and cost.

The rest of the semester